(Q1) 1) Yes, $\alpha(t)$ & $\beta(t)$ lie inside the feasible region. They must as the feasible region is independent of A_t (and hence, t as well). If they cross the boundary, something is wrong with the simulation.

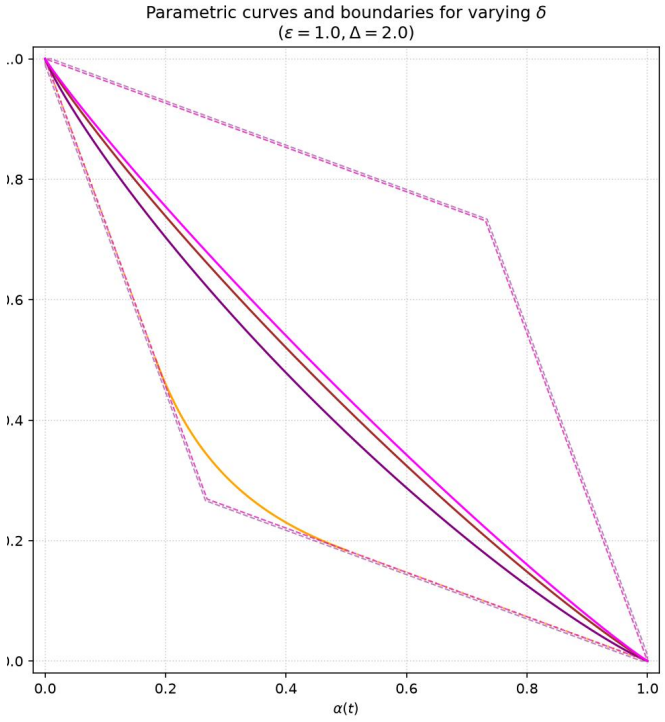
2) Laplace is much closer to boundary than gaussian, for all $\alpha \in [0,1]$.

$$\sigma_{Lap} = \sqrt{2}\lambda = \frac{\sqrt{2}\Delta}{\varepsilon}, \quad \sigma_{gauss} = \frac{\sqrt{2}\Delta}{\varepsilon}\sqrt{\ln\left(\frac{2}{\delta}\right)}$$

for $\delta < \frac{2}{e}$ (in our case mostly it is),

$$\Rightarrow \sqrt{\ln\left(\frac{2}{\delta}\right)} < 1 \Rightarrow \sigma_{gauss} > \sigma_{Lap}$$ for same privacy. $\Rightarrow$ Gauss mechanism adds much more noise, which makes it go away from boundary.

3)



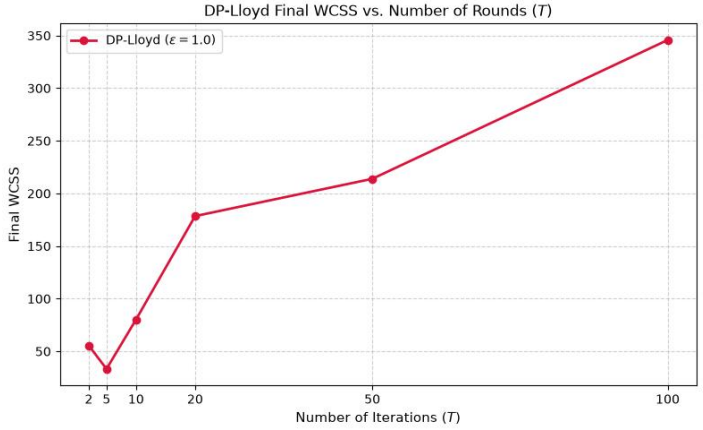
As $\delta \downarrow$, the gauss curves go far away from boundary and goes closer to $\alpha+\beta=1$ line which comes from σ_{gauss} increasing as $\delta \downarrow$.

Laplace curve does not move as σ_{Lap} doesn't depend on δ.

(Q2) 1) t= 5 | Lloyd WCSS: 49.3247 | DP-Lloyd WCSS: 225.3439
t=10 | Lloyd WCSS: 49.2993 | DP-Lloyd WCSS: 272.9774
t=15 | Lloyd WCSS: 49.2880 | DP-Lloyd WCSS: 115.1670
t=20 | Lloyd WCSS: 49.2862 | DP-Lloyd WCSS: 178.5032

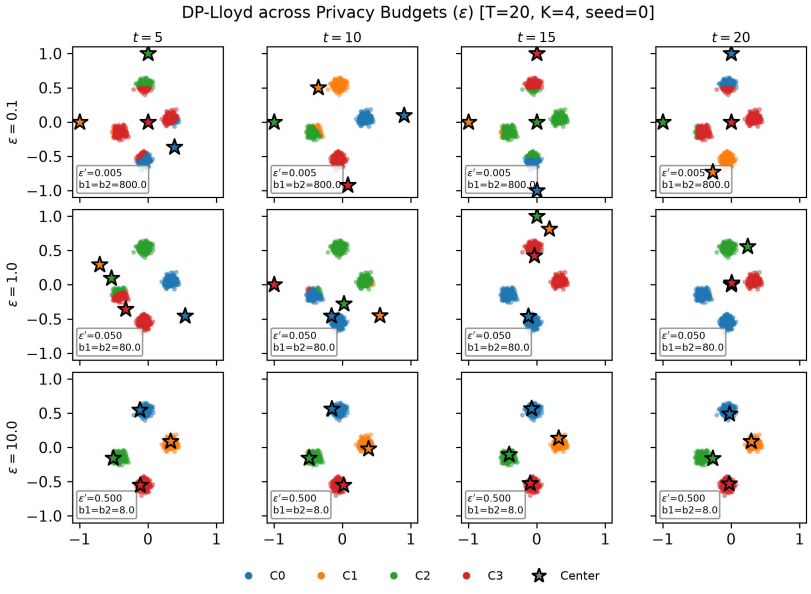
WCSS for Lloyd is monotonically non-increasing but WCSS of DP-Lloyd is not. The difference comes from the Laplace noise added which skews the monotonic behavior.

2)



as T↑, the WCSS tends to increase
This occurs as noise var $\left[b = \frac{\Delta}{\varepsilon'} = \frac{4T}{\varepsilon} \right]$ increases as T increases

3)



as $\varepsilon\uparrow$, $b = \frac{4T}{\varepsilon}\downarrow$
$\Rightarrow$ noise var decreases
so as $\varepsilon\uparrow$ the bottom row plots are closer to actual center means (higher utility)